

# METIMMOX-1 Model Input Parameters

Base Case, Deterministic Sensitivity Analysis (DSA), and Probabilistic Sensitivity Analysis (PSA) Parameters

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## 1 Summary

This report provides an overview of all input parameters used in the METIMMOX-1 health economic model, organized into global model settings and variable parameters across base case, DSA, and PSA analyses.

## 2 Global Model Parameters

These parameters define the overall model structure and are held constant across all analyses.

Table 1: Global Model Parameters

| Parameter           | Value                | Description                                | Source                                                                                               |
|---------------------|----------------------|--------------------------------------------|------------------------------------------------------------------------------------------------------|
| Time Horizon        | 520 weeks (10 years) | Model time horizon                         | Clinical follow-up period                                                                            |
| Cycle Length        | 1 week               | Cycle length for discrete-time model       | Model assumption                                                                                     |
| Discount Rate       | 4% per annum         | Annual discount rate for costs and effects | Model specification                                                                                  |
| WTP Threshold       | €51,000 per QALY     | Willingness-to-pay threshold               | Norwegian threshold based on severity (14.5 QALYs lost for diagnosis, based on a health 65-year old) |
| PSA Sample Size     | 5,000 iterations     | Number of Monte Carlo iterations for PSA   | Computational adequacy                                                                               |
| Bootstrap Samples   | 5,000 samples        | Bootstrap samples for survival uncertainty | Statistical adequacy                                                                                 |
| DSA Variation Range | ±20%                 | Parameter variation range for DSA          | Standard practice                                                                                    |

### 3 Model Input Parameters

This table presents all variable model parameters organized by type, showing their base case values, DSA variation ranges, and PSA distribution specifications.

Table 2: Model Input Parameters by Type

| Type                         | Parameter                             | Base Case                         | DSA Range         | PSA Distribution                   |
|------------------------------|---------------------------------------|-----------------------------------|-------------------|------------------------------------|
| <b>Cost Parameters</b>       |                                       |                                   |                   |                                    |
| Cost                         | Nivolumab per administration          | €13,923                           | €11,138 - €16,708 | Gamma (shape=25, rate=0.0018)      |
| Cost                         | FLOX per administration               | €427                              | €342 - €512       | Gamma (shape=25, rate=0.0586)      |
| Cost                         | CT scan                               | €386                              | €309 - €463       | Gamma (shape=25, rate=0.0648)      |
| Cost                         | Blood tests                           | €0                                | Not varied        | Fixed at 0                         |
| Cost                         | NGS testing                           | €1,439                            | €1,151 - €1,727   | Gamma (shape=25, rate=0.0174)      |
| Cost                         | Standard outpatient visit             | €33                               | Not varied        | Gamma (shape=25, rate=0.7576)      |
| Cost                         | Baseline visit                        | €530                              | Not varied        | Gamma (shape=25, rate=0.0472)      |
| Cost                         | Follow-up visit                       | €33                               | Not varied        | Gamma (shape=25, rate=0.7576)      |
| Cost                         | End-of-life care                      | €13,803                           | €11,042 - €16,564 | Gamma (shape=25, rate=0.0018)      |
| <b>Utility Parameters</b>    |                                       |                                   |                   |                                    |
| Utility                      | Non-progressed state                  | 0.9077                            | 0.7262 - 1.0000   | Beta ( =41.0, =4.2)                |
| Utility                      | Progressed state                      | 0.9005                            | 0.7204 - 1.0000   | Beta ( =40.6, =4.5)                |
| <b>Prevalence Parameters</b> |                                       |                                   |                   |                                    |
| Prevalence                   | CRP biomarker positive                | 30%                               | Not varied        | Fixed                              |
| Prevalence                   | TLR biomarker positive                | 25%                               | Not varied        | Fixed                              |
| Prevalence                   | TMB/BRAF biomarker positive           | 20%                               | Not varied        | Fixed                              |
| <b>Survival Parameters</b>   |                                       |                                   |                   |                                    |
| Survival                     | OS and PFS curves by biomarker status | Fitted parametric survival models | Not varied        | Bootstrap sampling (5,000 samples) |

*Notes:*

DSA ranges calculated as base case  $\pm$  20%. Utility upper bounds capped at 1.0.

PSA: Gamma distributions for costs (CV=20%), Beta for utilities (CV=15%).

Survival curves use bootstrap sampling with age and sex adjustments.

## 4 PSA Parameter Distributions

This section displays histograms of all sampled parameters from the probabilistic sensitivity analysis (PSA), showing the distribution shapes and ranges used in the Monte Carlo simulation.

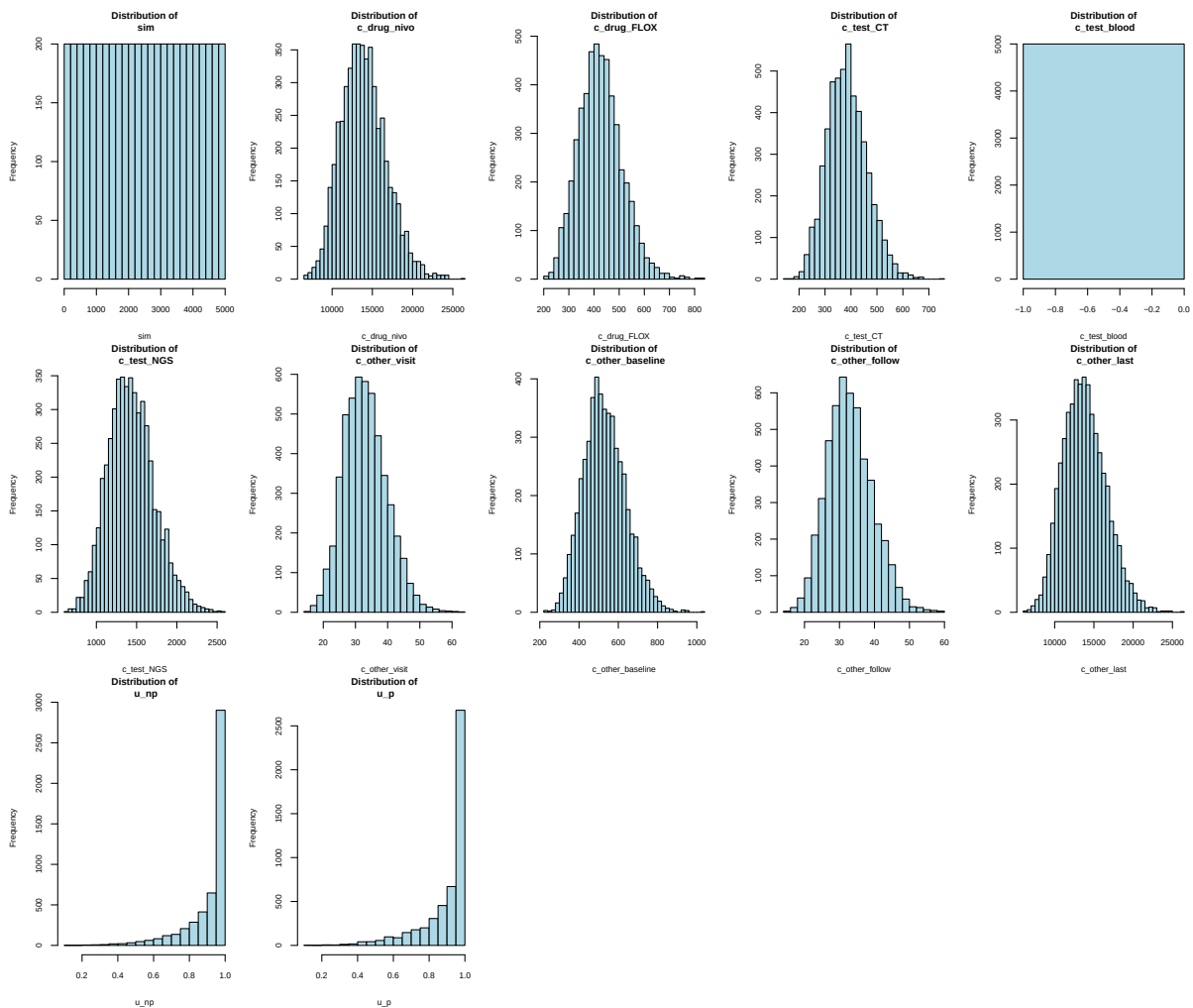


Figure 1: Distribution of PSA Parameters

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